

Test Cases Document

This document outlines the comprehensive test suite implemented for the Flaky Test Detector & Explainer project.

1. Universal Data Parser Tests (test_universal_parser.py)

- test_original_sample_format – Standard CSV parsing and normalization.
- test_jenkins_report_format – Jenkins CSV schema inference and warning generation.
- test_github_actions_format – GitHub Actions JSON parsing and normalization.
- test_excel_spreadsheet – Excel file ingestion and extraction.
- test_missing_columns_graceful_handling – Default values and warning generation.
- test_ai_heuristics_inference – Column inference using data heuristics.
- test_empty_file – Empty file validation and ValueError handling.

2. Core Detector Logic Tests (test_detector.py)

- test_load_test_runs_success – Successful CSV loading and mapping.
- test_load_test_runs_missing_file – FileNotFoundError validation.
- test_calculate_test_statistics – Statistics aggregation verification.
- test_detect_flaky_tests – Flaky test identification and ranking.
- test_severity_classification – LOW, MEDIUM, HIGH severity assignment.
- test_analyze_test_trends – Trend and failure trajectory analysis.

3. Web & API Route Tests

- /api/upload – Upload, parse, and storage workflow validation.
- /api/flaky-tests – Dashboard JSON payload validation.
- /api/chat – Ollama and Groq fallback integration validation.

Coverage Summary

The suite validates parser resilience, detector accuracy, severity classification, trend analysis, API functionality, and AI integration behavior.